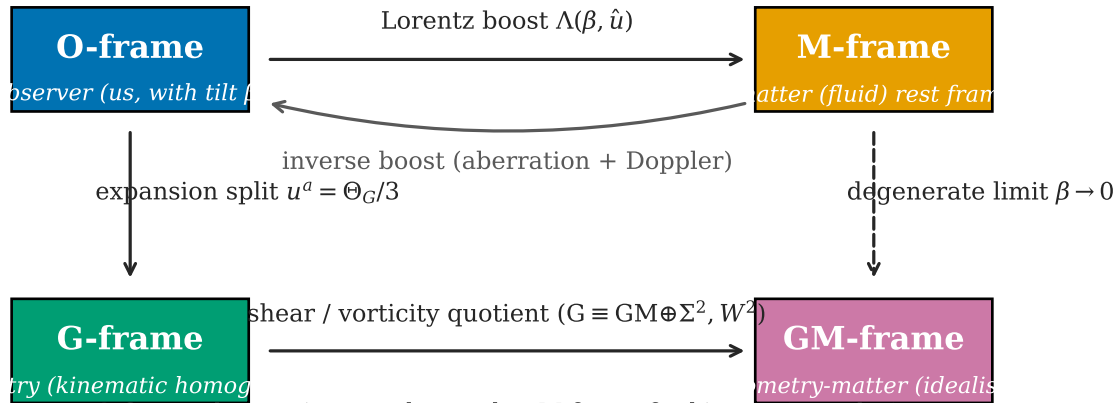


Four-frame tetrad decomposition (ch03 §frames)



O-frame observations are boosted to M-frame for kinematic analysis;

shear Σ^2 and vorticity W^2 are the $G \rightarrow GM$ quotient charge.

GM-frame is the hypothetical FLRW limit ($\Sigma^2 = W^2 = 0$); distance to GM is the defect.